

TRI-TIERED LOCAL LLM ACTIVE LEARNING FRAMEWORK

Empirical Evaluation & Cost-Efficient Routing Report

Executive Research Summary

This research report compiles results from the empirical evaluation of the Tri-Tiered Active Learning Router. The tiered framework dynamically routes incoming text classification queries to appropriate computation layers (Autonomous DistilBERT, Assisted Qwen2.5:3B LLM, or Human Expert Annotator) based on uncertainty and entropy thresholds. By leveraging dynamic validation threshold sweeps and experience replay active learning, the framework achieves significant reductions in human expert workloads while maximizing model accuracy and utility.

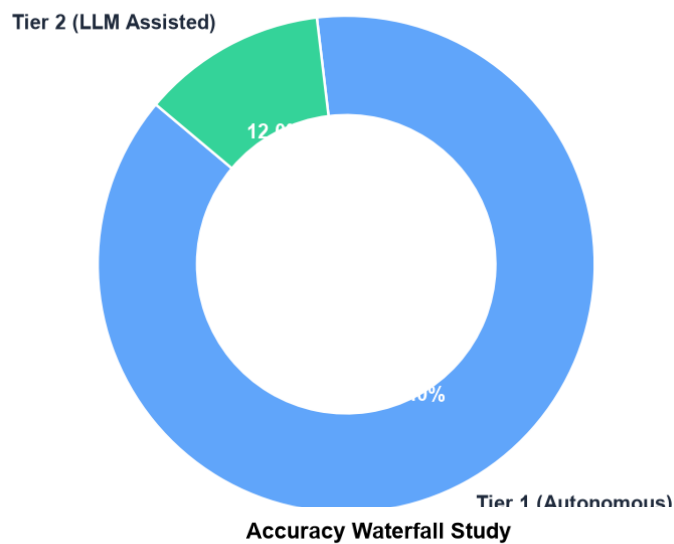
Dataset	T1 Acc (Pre)	T1 Acc (Post)	Final Acc	Human Effort %	Net Util (Human)	Net Util (Cost)	Avg Cost
AG_NEWS	87.40%	86.20%	86.80%	0.00%	0.0060	0.0056	24.88
DBPEDIA_14	97.60%	96.40%	97.80%	0.30%	0.0139	0.0136	24.13
EMOTION	58.80%	59.60%	62.60%	0.40%	0.0298	0.0269	184.53
IMDB	79.40%	78.20%	87.60%	2.40%	0.0928	0.0914	157.75

Dataset Analysis: AG_NEWS

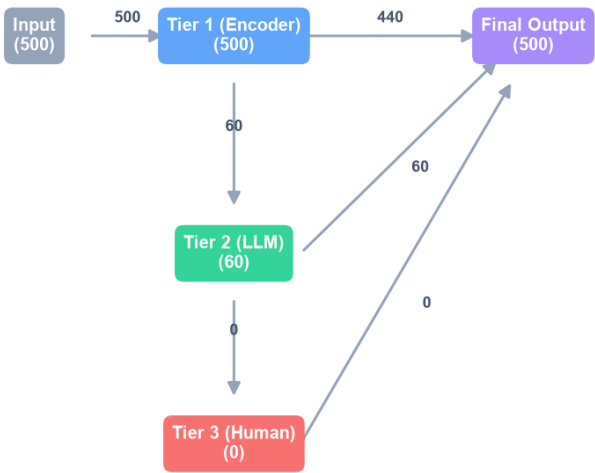
Performance & Budget Summary Metrics

Total Evaluated Samples:	500	Human Effort Ratio:	0.00%
Pre-AL T1 Accuracy:	87.40%	Post-AL T1 Accuracy:	86.20%
Routed Final System Accuracy:	86.80%	Final System F1 Macro:	0.8705
Brier Score (Final):	0.2255	Final System ECE:	0.0778
Net Utility (Human-Effort):	0.0060	Net Utility (Cost-Aware):	0.0056
Average Compute Cost:	24.88	Active Learning Escalations:	5 sample(s)

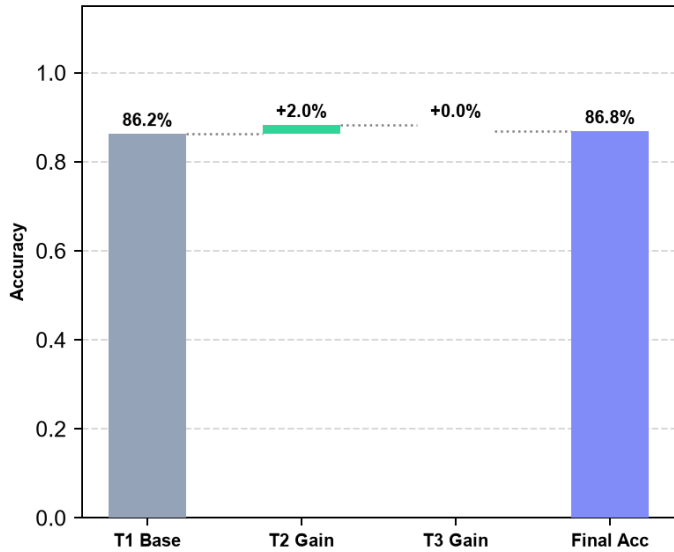
Request Routing Distribution



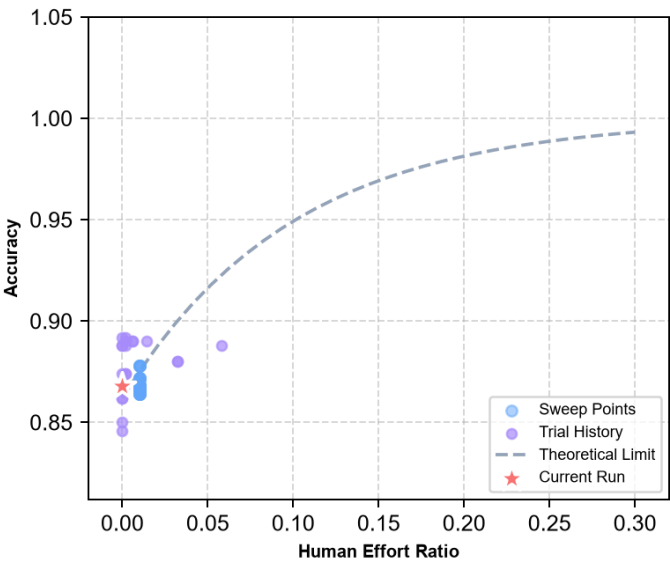
Request Flow Pipeline (Sankey)



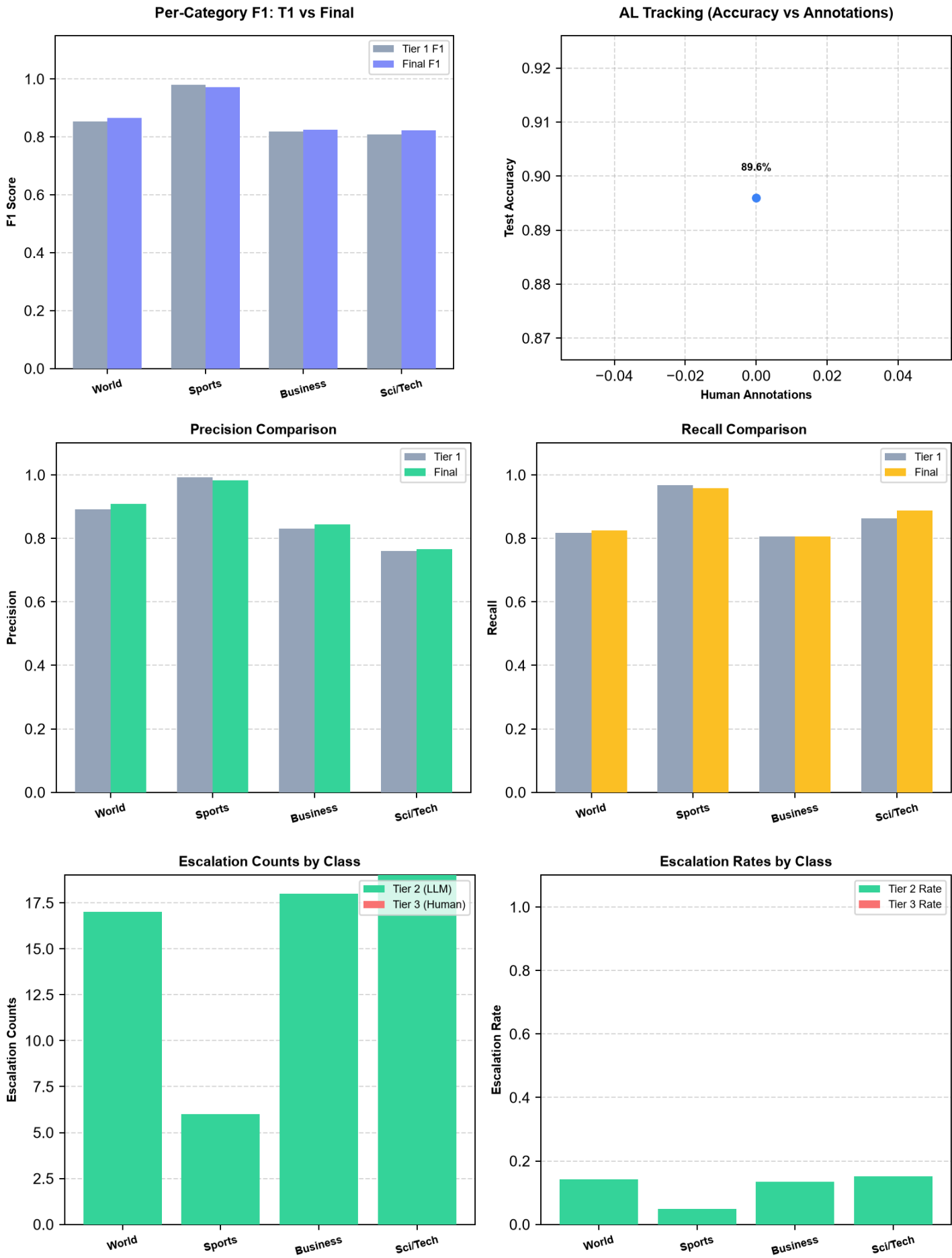
Accuracy Waterfall Study



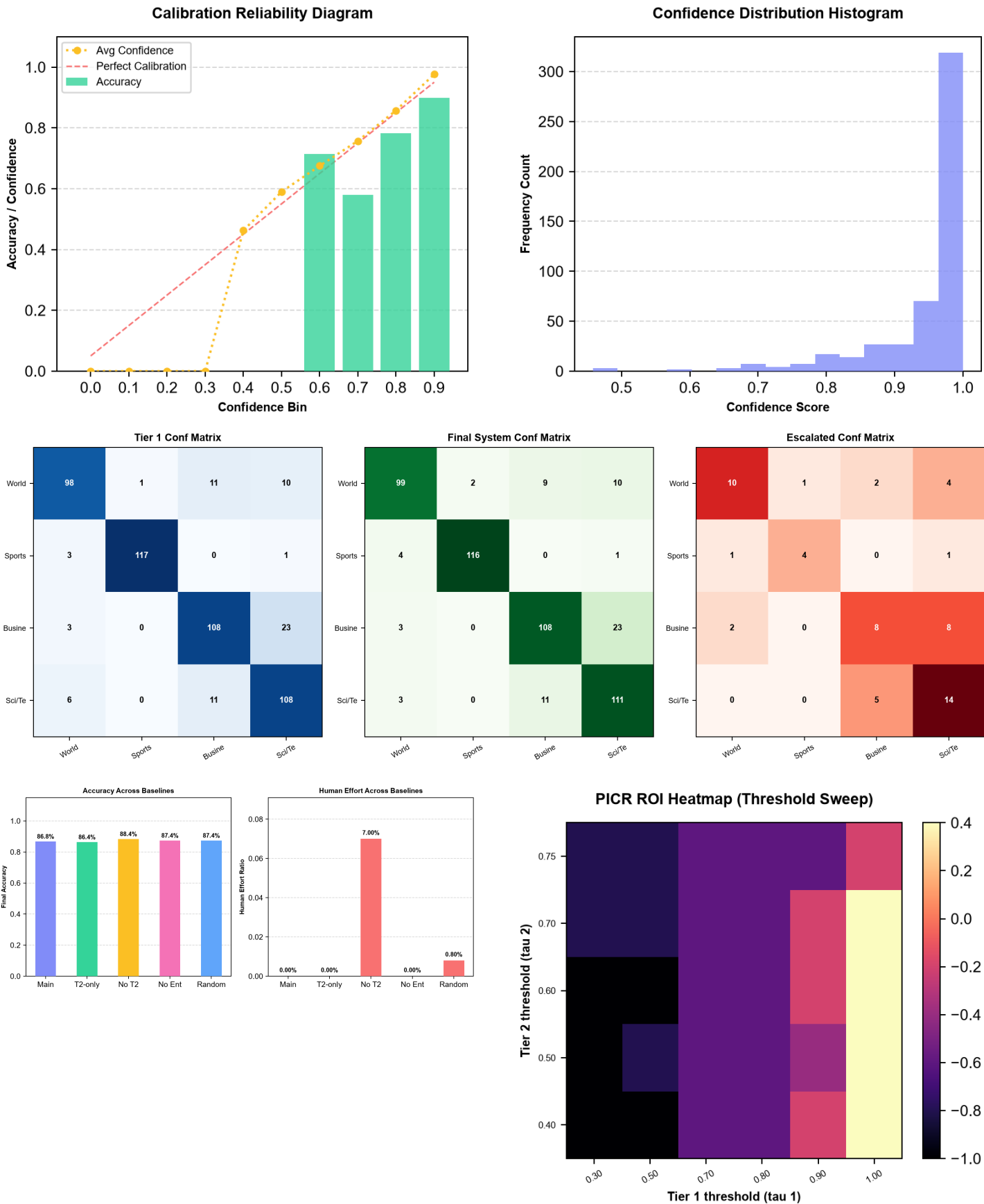
Efficiency Frontier Analysis



Category & Escalation Profile: AG_NEWS



Calibration & Validation Studies: AG_NEWS

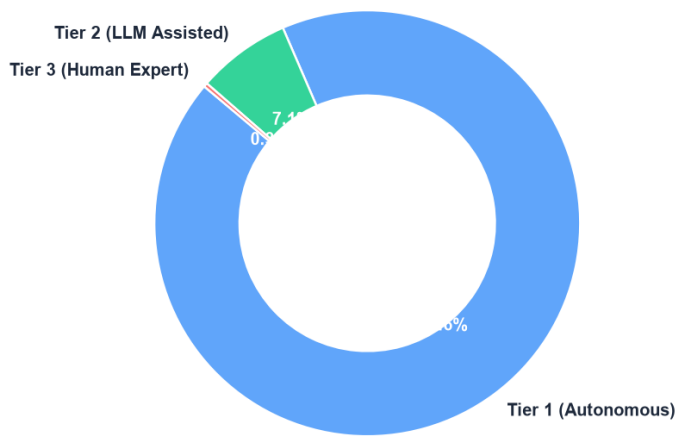


Dataset Analysis: DBPEDIA_14

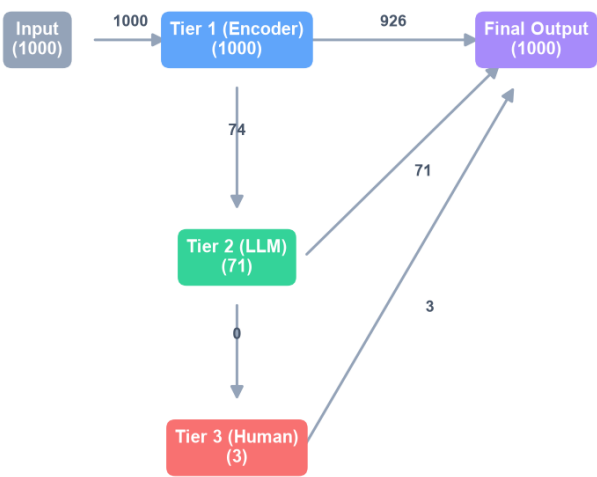
Performance & Budget Summary Metrics

Total Evaluated Samples:	1000	Human Effort Ratio:	0.30%
Pre-AL T1 Accuracy:	97.60%	Post-AL T1 Accuracy:	96.40%
Routed Final System Accuracy:	97.80%	Final System F1 Macro:	0.9790
Brier Score (Final):	0.0440	Final System ECE:	0.0156
Net Utility (Human-Effort):	0.0139	Net Utility (Cost-Aware):	0.0136
Average Compute Cost:	24.13	Active Learning Escalations:	3 sample(s)

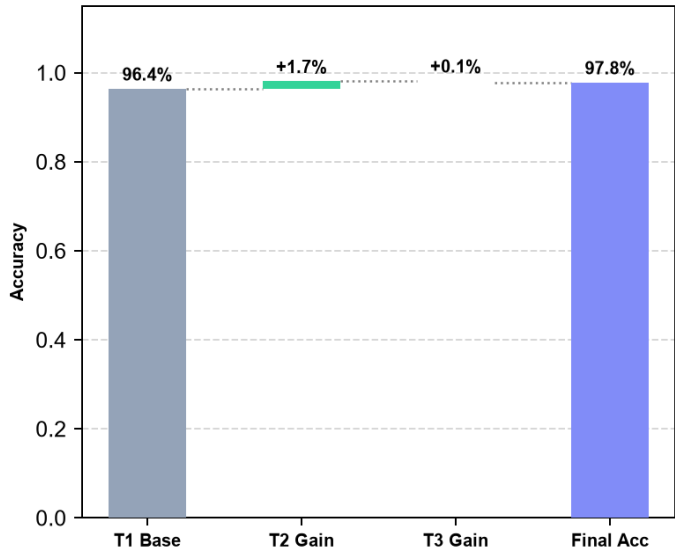
Request Routing Distribution



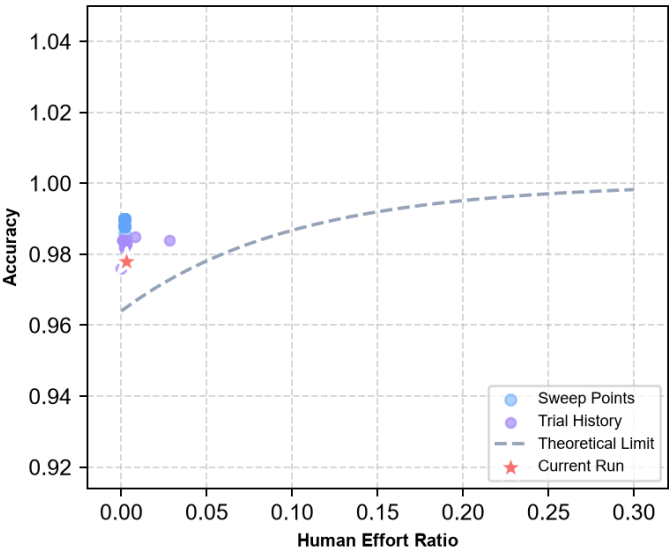
Request Flow Pipeline (Sankey)



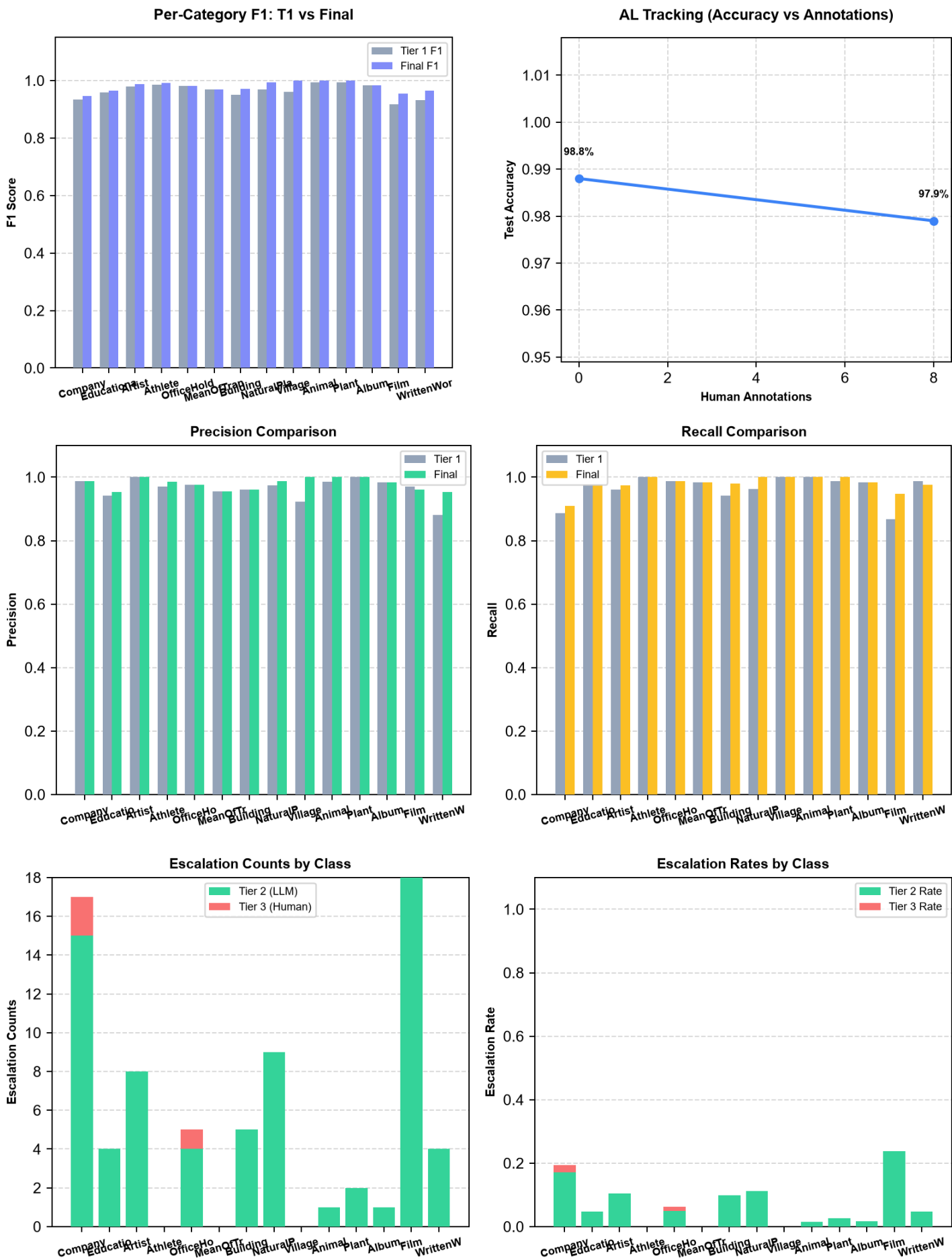
Accuracy Waterfall Study



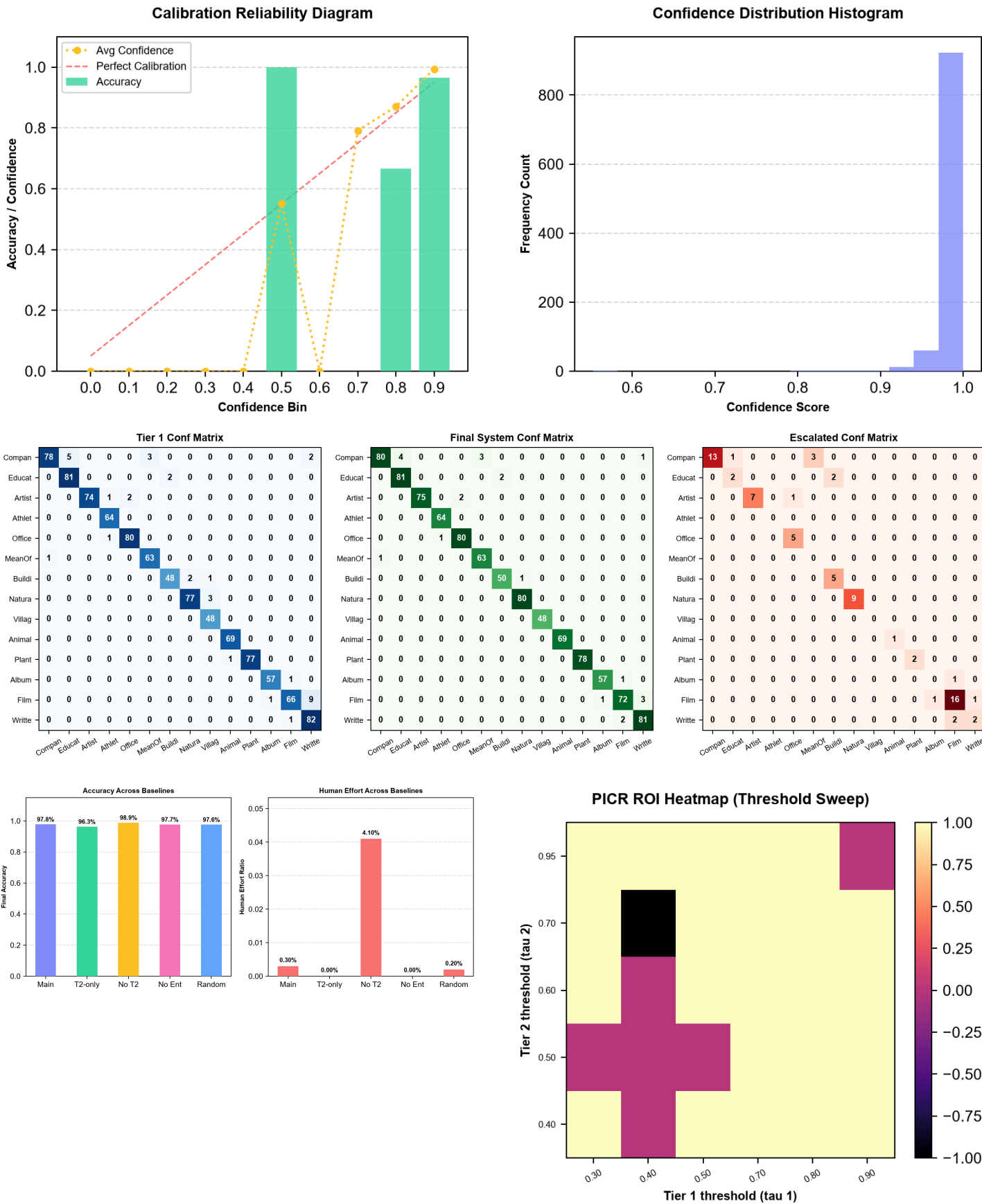
Efficiency Frontier Analysis



Category & Escalation Profile: DBPEDIA_14



Calibration & Validation Studies: DBPEDIA_14

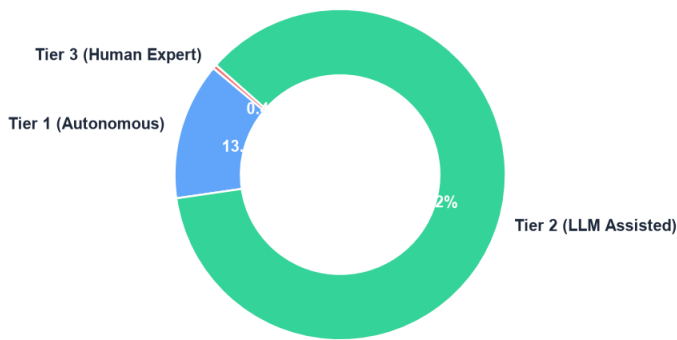


Dataset Analysis: EMOTION

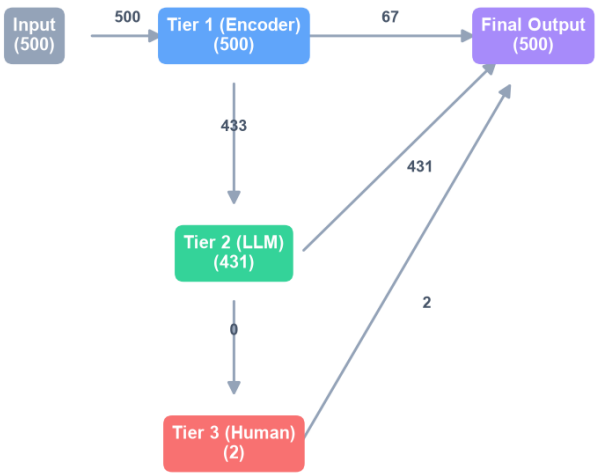
Performance & Budget Summary Metrics

Total Evaluated Samples:	500	Human Effort Ratio:	0.40%
Pre-AL T1 Accuracy:	58.80%	Post-AL T1 Accuracy:	59.60%
Routed Final System Accuracy:	62.60%	Final System F1 Macro:	0.4303
Brier Score (Final):	0.6239	Final System ECE:	0.2355
Net Utility (Human-Effort):	0.0298	Net Utility (Cost-Aware):	0.0269
Average Compute Cost:	184.53	Active Learning Escalations:	7 sample(s)

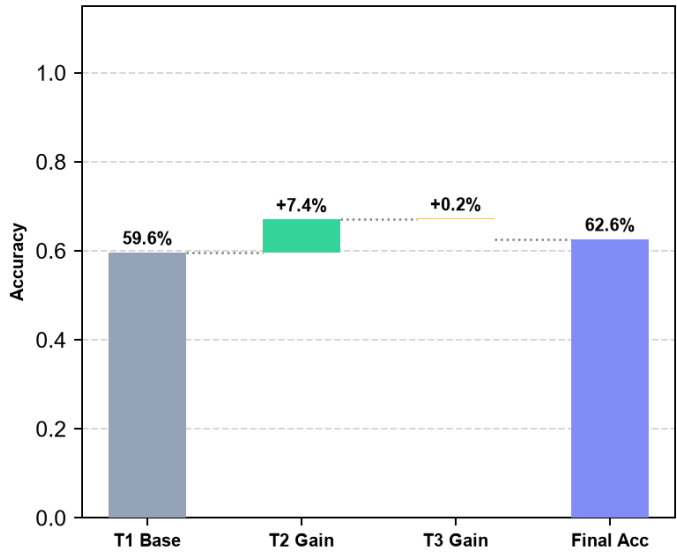
Request Routing Distribution



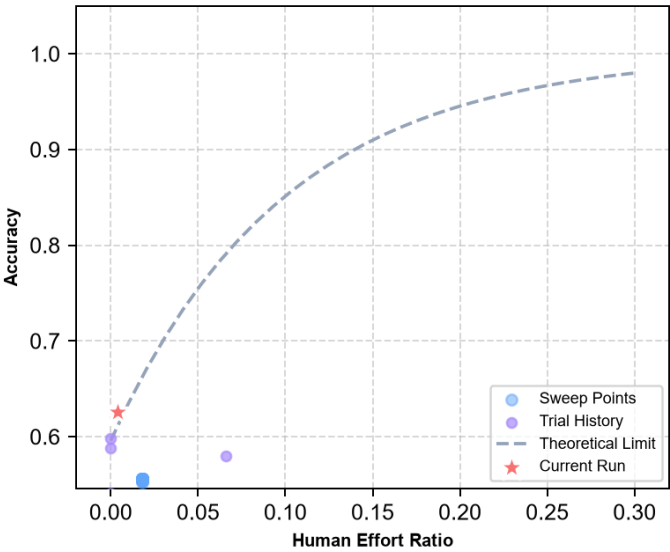
Request Flow Pipeline (Sankey)



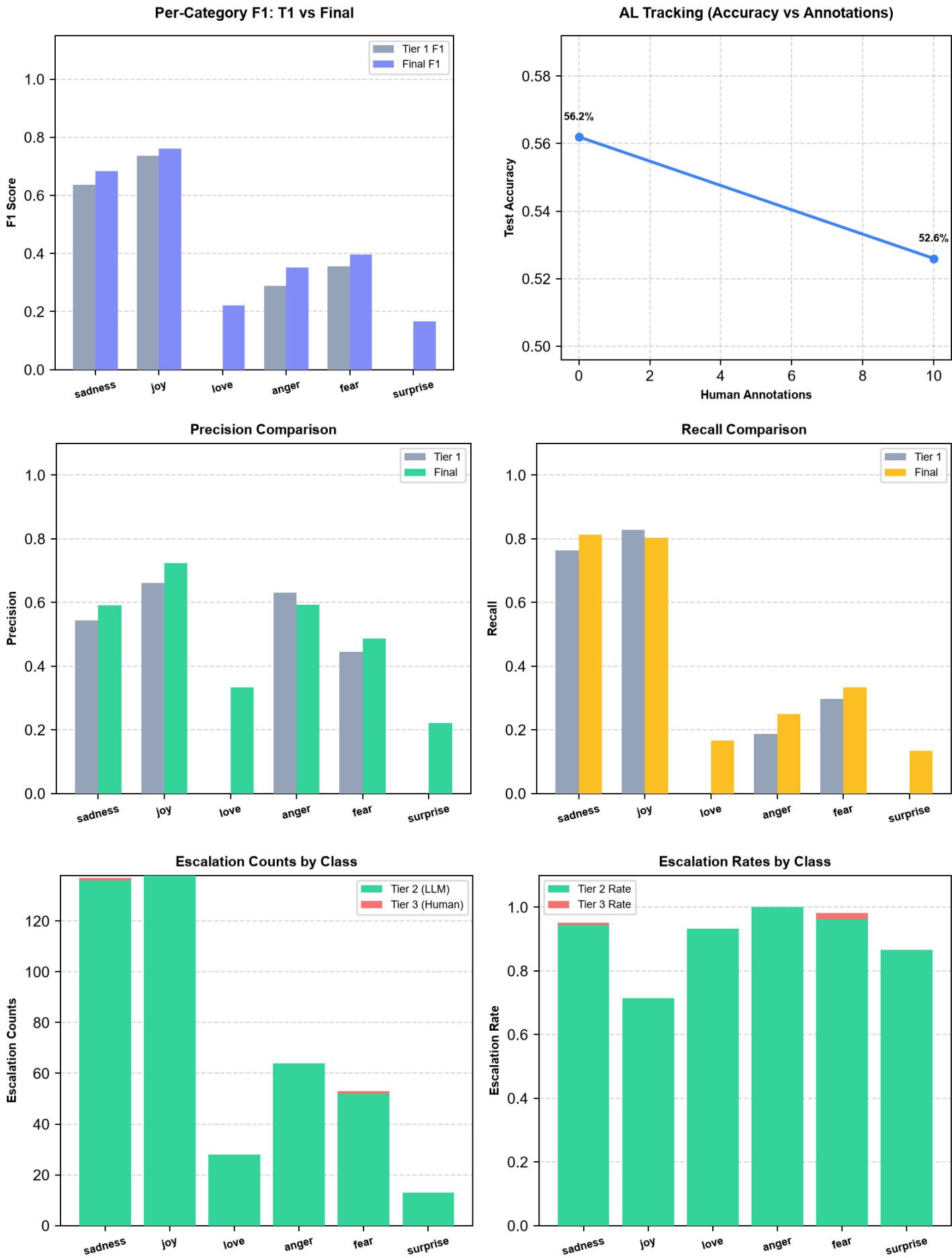
Accuracy Waterfall Study



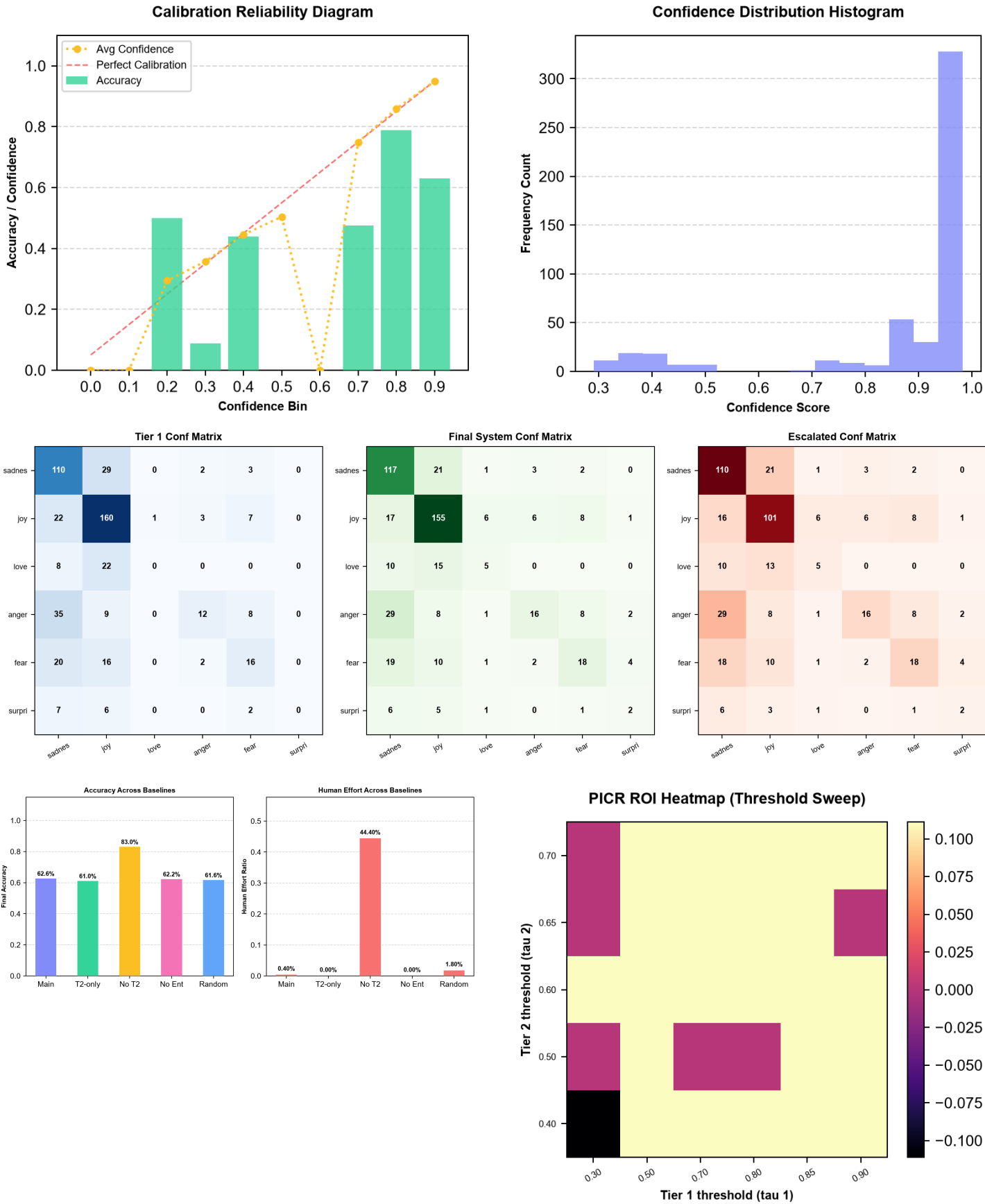
Efficiency Frontier Analysis



Category & Escalation Profile: EMOTION



Calibration & Validation Studies: EMOTION

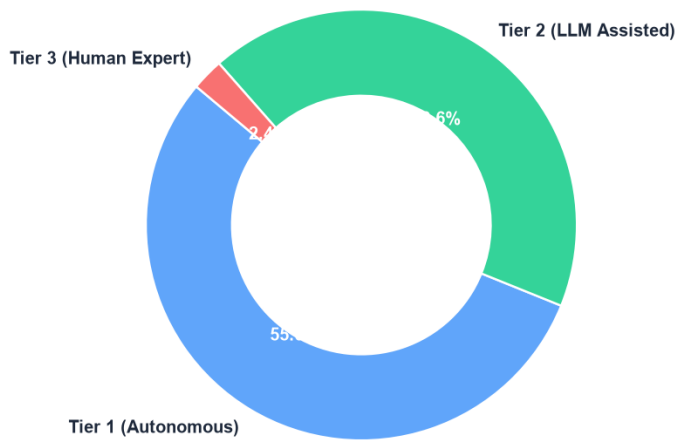


Dataset Analysis: IMDB

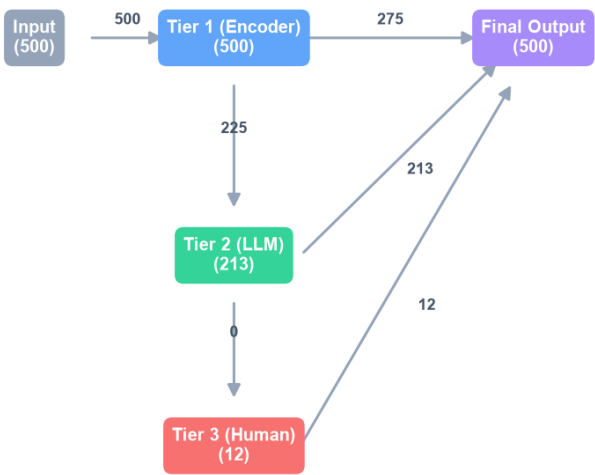
Performance & Budget Summary Metrics

Total Evaluated Samples:	500	Human Effort Ratio:	2.40%
Pre-AL T1 Accuracy:	79.40%	Post-AL T1 Accuracy:	78.20%
Routed Final System Accuracy:	87.60%	Final System F1 Macro:	0.8749
Brier Score (Final):	0.2333	Final System ECE:	0.0664
Net Utility (Human-Effort):	0.0928	Net Utility (Cost-Aware):	0.0914
Average Compute Cost:	157.75	Active Learning Escalations:	24 sample(s)

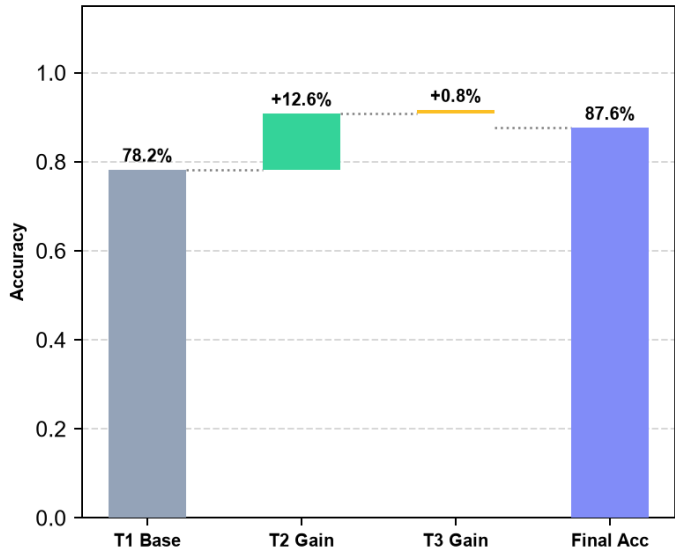
Request Routing Distribution



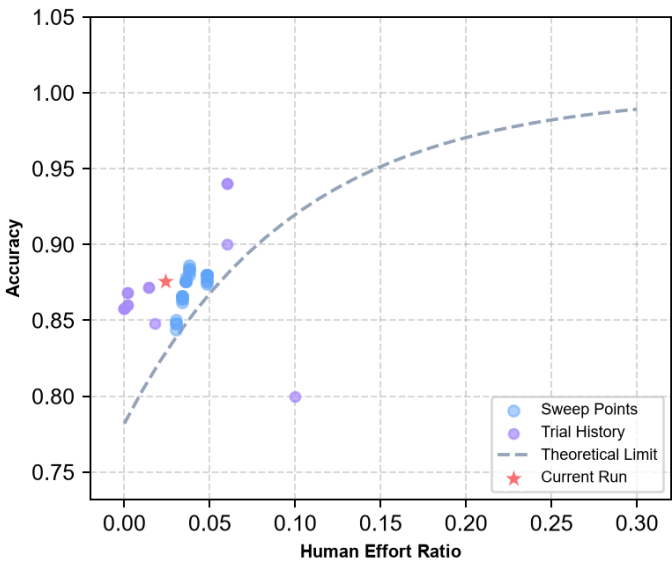
Request Flow Pipeline (Sankey)



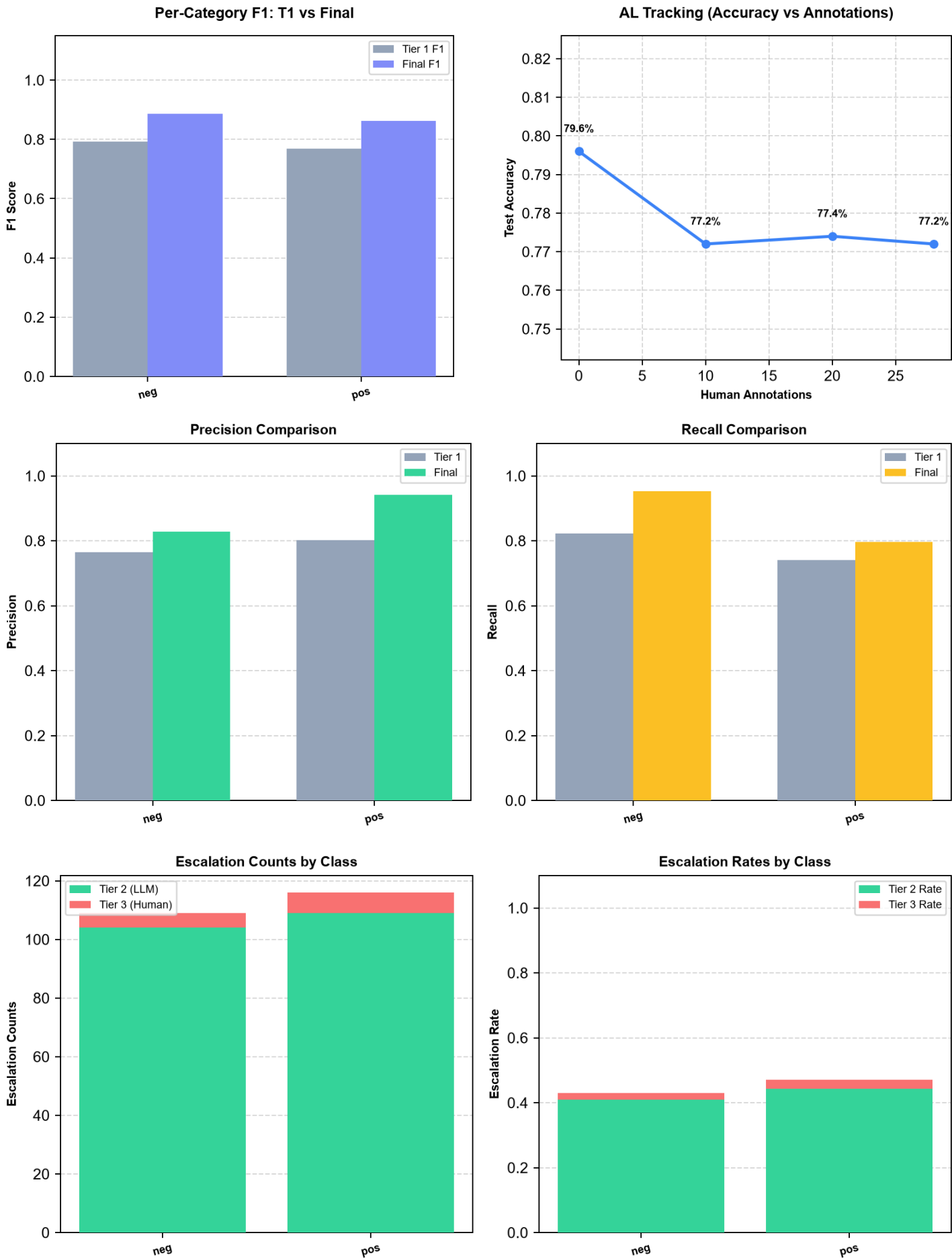
Accuracy Waterfall Study



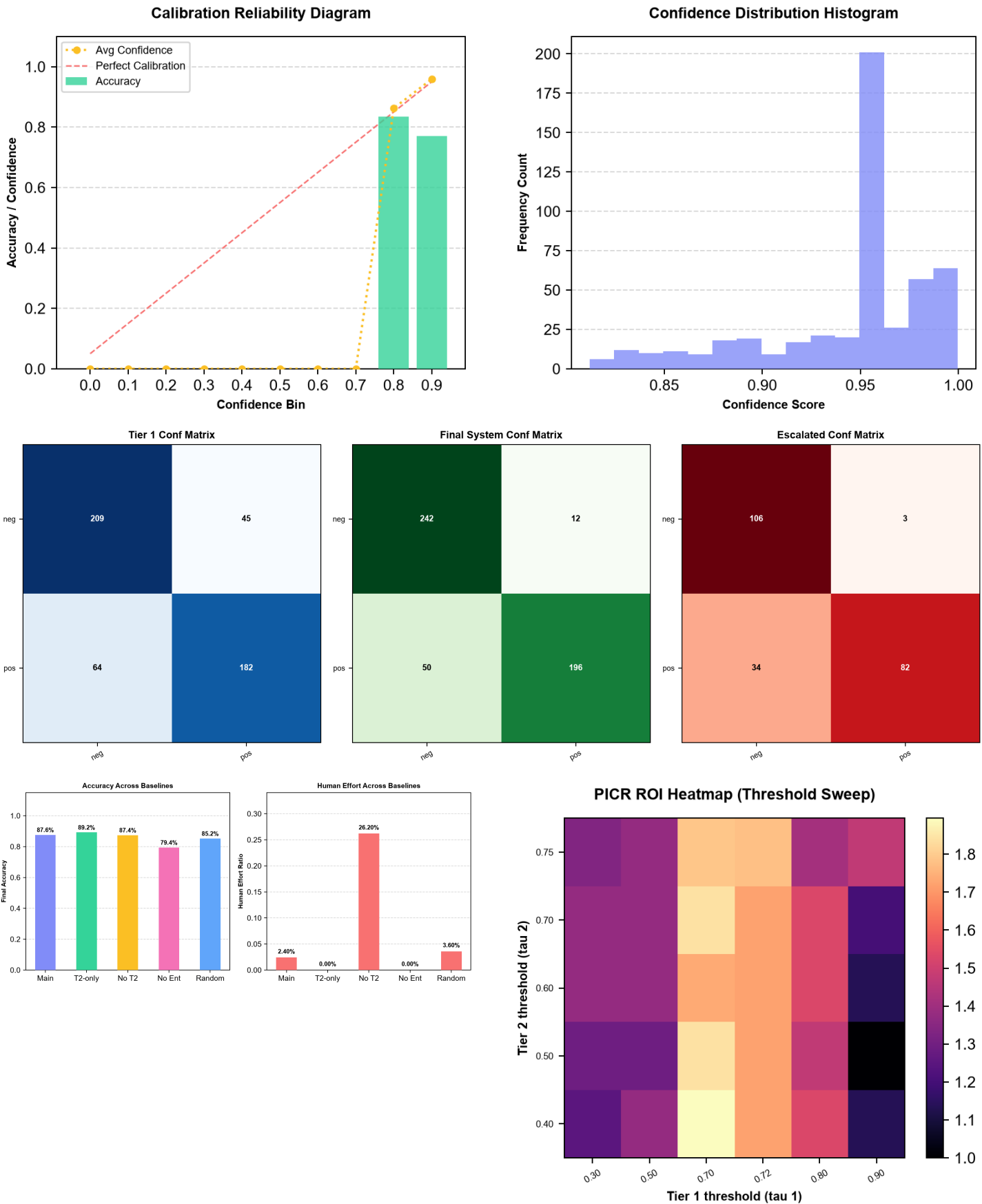
Efficiency Frontier Analysis



Category & Escalation Profile: IMDB



Calibration & Validation Studies: IMDB



Cross-Dataset Comparison Synthesis

The benefit of adaptive routing varies substantially across datasets. IMDb exhibits the largest gain from LLM-assisted routing, while DBpedia-14 shows smaller gains because its Tier-1 classifier is already highly accurate. Emotion represents a challenging case where routing and adaptation do not recover the strong pre-AL Tier-1 performance, highlighting the dependence of the framework on base-model calibration and task characteristics.

